Adsorption of Amino Acids onto Hydroxyapatite

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Adsorption isotherms at 37°C were obtained for L-aspartic, L-glutamic, DL-α-amino-adipic, DL-α-amino-pimelic, succinic, and adipic acids, and L-arginine onto hydroxyapatite. A Langmuir-type adsorption model fitted the experimental results. The values of the derived adsorption parameters indicate that the nature of the adsorption bond is the same for the four acidic amino acids. The results are consistent with an adsorption mechanism in which terminal carboxyl groups are attached to calcium ions on the surface of the adsorbent. The observed differences in the maximum number of adsorption sites are explained by a geometric model based on the size of the adsorbate molecules and the distances between calcium ions obtained from the crystallographic structure of hydroxyapatite. The geometric model also fits the adsorption data for the two dicarboxylic acids although these molecules display a higher adsorption affinity for the hydroxyapatite surface than the four acidic amino acids. No adsorption of L-arginine was detected unless the hydroxyapatite was pretreated with a phosphate solution. The pretreatment was more effective when the pH of the phosphate solution was 7.8 than when the pH was 6.8.

The difference in adsorption affinities of organic molecules for hydroxyapatite, HA, $Ca_5OH(PO_4)_3$, is the basis for the use of this calcium phosphate in chromatographic separations of complex compounds such as serum proteins (1, 2). The interaction of HA with organic molecules is also of biological importance, since the former is considered as the prototype mineral found in calcified tissues such as bone and teeth, where intimate associations have been reported (3, 4) between the mineral and organic phases. Recently, for example, it was reported (5) that the process of adsorption of salivary glycoproteins onto tooth surfaces is selective and that the dental pellicle so formed confers a remarkable protection against acid demineralization of the tooth enamel (6). However, in spite of its chromatographic and biological significance, little is known about the mechanism of adsorption of proteins onto HA.

Elucidation of the adsorption mechanisms of natural macromolecules onto HA is hindered by the complexity of the molecular structures and diversity of functional groups usually found in such compounds. It seems more appropriate to use, as an initial step, small molecules with a restricted number of functional groups. We report here the results of adsorption studies of simple α-amino and dicarboxylic acids. Although there are reports (7, 8) on this subject, the present paper emphasizes quantitative aspects obtained from an analysis of adsorption isotherms. Parameters derived from these isotherms are used to advance a mechanistic view of the adsorption phenomenon.

MATERIALS AND METHODS

Hydroxyapatite. Most of the present work was conducted using a commercially available HA (Monsanto) which will be referred to as HA–M. The X-ray pattern of this preparation displayed only the peaks characteristic for HA although marked broadening was apparent due to its small particle size. Under the polarizing microscope HA–M appears as a homogeneous microcrystalline phase with a mean index of refraction of 1.632 ± 0.002. Chemical analyses

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TABLE I

Adsorption Isotherm of L-Aspartic Acid on HA–M at 37°C[a]

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
4.51	0.300	3.02
6.01	0.396	4.11
10.5	0.712	6.81
15.0	1.05	9.11
21.0	1.49	12.2
30.1	2.20	16.0
37.6	2.75	20.1
45.1	3.43	21.6
26.3	3.20	20.6
30.0	2.18	16.5
37.5	2.77	19.6
60.1*	4.64	27.4
75.1*	5.86	33.0

[a] Entries excluded from linear regression of Eq. [1] are indicated by asterisks.

yielded contents of 39.1 ± 1.0% Ca and 18.1 ± 0.3% P with a Ca/P molar ratio of 1.67 ±0.02. The specific surface area, determined by nitrogen adsorption, was 59.56 m^2/g.

Another HA preparation, HA–L, was obtained by a precipitation procedure previously described (9). The X-ray pattern of this HA displayed only narrow peaks corresponding to the reflections of a well crystallized HA. Its specific surface area was 13.55 m^2/g. Under the microscope, HA–L appeared as an acicular crystalline phase with a mean refraction index of 1.636 ± 0.002. The Ca and P contents were 39.2% and 18.0%, respectively, yielding a Ca/P molar ratio of 1.68 ± 0.02.

Adsorbates. All the compounds used were reagent grade and included: L-aspartic (Asp), L-glutamic (Glu), DL-α-amino-adipic, DL-α-amino-pimelic, succinic, and adipic-acids, and L-arginine. In some experiments, L-Glu (^{14}C) and L-Asp (^{3}H) were used to check the accuracy of the conventional analytical methods. Because of the low specific surface area of HA–L, the adsorption isotherm for Asp on this material was obtained using solutions tagged with L-Asp (^{3}H).

Equilibrations. Accurately weighed samples of HA (100–400 mg) were placed in stoppered plastic tubes containing 2 ml of adsorbate solution at various concentrations (from about 20 to 1000 μg/ml). Control tubes containing only the solution were included in all experiments; no adsorption was detected on to the surfaces of the plastic-ware used in the preparation or analyses of the solutions. The tubes were mounted on an end-over-end shaker and the equilibration was conducted at 37 ± 0.5°C for 1 hr (this time was selected after preliminary experiments showed invariant concentrations for longer equilibration times). The tubes were then centrifuged at the same temperature, at 4000*g* (max) for 15 min. Aliquots of the supernatant were taken for chemical analyses. The amounts adsorbed were obtained as differences between the initial and final adsorbate concentrations. Plots of the amount adsorbed per gram of HA, Q, versus equilibrium concentrations, C_f, constitute the adsorption isotherms for the various compounds under consideration.

The initial conditions for each adsorption experiment are reported as a combined figure, R, which is the initial concentration per gram of HA used. The actual weight, W, of HA and the initial concentration of adsorbate in solution, C_i, can be calculated from $W = C_f V/(RV - Q)$ and $C_i = WR$. The solution volume V was 2 ml in all experiments.

TABLE II

Adsorption Isotherm of L-Glutamic Acid on HA–M at 37°C[a]

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
1.7	0.230	1.10
3.4	0.473	2.08
6.8	0.974	3.85
10.2	1.44	6.03
13.6	1.97	7.53
17.0	2.55	8.44
20.4	3.15	9.31
23.8	3.71	10.5
27.2*	4.22	12.1
30.6*	4.77	13.4
34.0*	5.19	16.1

[a] Entries excluded from linear regression of Eq. [1] are indicated by asterisks.

Analytical. Amino acid concentrations were determined by the indanetrione hydrate method (10, 11). Excellent agreement was found between adsorption figures for Asp and Glu obtained by this method and those obtained by the use of solutions tagged with Glu (^{14}C) and Asp (^{3}H). In the latter case, the analyses were performed using a liquid scintillation counter; the "cocktail" used was a solution containing 1400 ml of toulene, 600 ml of absolute ethanol, 8 g of PPO, and 200 mg of POPOP. To 15 ml of "cocktail" was added 0.5 ml of solution containing the adsorbate and 0.2 ml of Protosol (New England Nuclear) to increase miscibility. The adsorption of dicarboxylic acids was determined using solutions of these compounds containing ^{14}C-labeled molecules. Uncertainties in the adsorbate analyses by the chemical and radioactive methods were estimated at ± 3 and $\pm 5\%$ of the amounts analyzed, respectively.

The pH values of the adsorbate solutions were determined, prior to and after equilibration, with a glass–calomel combination electrode. For the acidic amino acids and dicarboxylic acids the initial solutions had pH values with the range 3.3–4.5; after equilibration the range was 5.3–6.8. For arginine, these ranges were 9.2–9.4 and 6.8–7.6, respectively. Buffers were not used in the present equilibrations, to prevent possible adsorption competition.

Pretreatments. In one set of experiments, an HA–M sample was suspended in 0.1 M $CaCl_2$ for 0.5 hr, centrifuged, and decanted. The same operation was repeated three times using 0.001 M $CaCl_2$, after which the solid was recovered and dried at 130°C overnight. This pretreated solid was then used to obtain an adsorption isotherm for Asp.

TABLE III

Adsorption Isotherm of DL-α-Amino-Adipic Acid on HA–M at 37°C

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
1.03	0.226	0.557
3.11	0.505	1.15
4.13	0.931	2.07
6.18	1.43	2.89
8.24	1.96	3.51
8.84	2.39	4.07

TABLE IV

Adsorption Isotherm of DL-α-Amino-Pimelic Acid on HA–M at 37°C

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
1.14	0.326	0.656
1.43	0.411	0.799
2.14	0.628	1.14
2.84	0.845	1.48
4.28	1.30	2.06
5.68	1.77	2.56
5.70	1.77	2.57
7.15	2.25	3.03

In another set of experiments, an HA–M sample was suspended in 0.1 M potassium phosphate at pH 7.8, centrifuged, and decanted. This operation was then repeated three times using 0.001 M potassium phosphate at the same pH (7.8). In a fashion similar to that with the $CaCl_2$ pretreatment, the recovered HA–M was used to obtain an adsorption isotherm for L-arginine. In this latter case, three additional adsorption experiments were conducted in the presence of the 0.001 M potassium phosphate.

Adsorption model. The adsorption isotherms reported here were obtained by direct measurements on the systems investigated. A general interpretation of the results, however, is best achieved by the use of an adsorption model. For the present investigation we have adopted a model that follows closely the Langmuir adsorption isotherm (12, 13). Thus, the experimental results should be described by

$$\frac{C}{Q} = \frac{1}{KN} + \frac{C}{N}. \qquad [1]$$

In Eq. [1], C is the solution concentration of adsorbate in equilibrium with the HA, Q and N are the amount adsorbed and the maximum number of adsorption sites per gram of adsorb-

TABLE V

Adsorption Isotherm of Succinic Acid on HA–M at 37°C

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
8.42	0.471	7.46
8.48	0.476	7.43
16.9	1.05	12.9
25.3	1.72	17.2
33.9	2.39	19.9
42.0	3.10	22.6

ent, respectively, the parameter K reflects the affinity that the adsorbate molecules have for the adsorption sites. It is apparent that a plot of the left-hand term of Eq. [1] versus C should be a straight line and from the values of the slope and intercept the parameters K and N can be calculated. In the present work, the slope and intercept values used for each isotherm were those obtained for the line of best fit according to a least-squares criterion. The correlation coefficient of the individual linear regressions are reported here to indicate the degree of agreement between Eq. [1] and the experimental data.

RESULTS

Tables I through VIII give the analytical figures used to obtain the adsorption isotherms for the various adsorbates used. The data for four amino acids (Tables I–VI) and two dicarboxylic acids (Tables V and VI) are plotted, according to Eq. [1], in Fig. 1. It is apparent that the plots for the four amino acids form a series of lines, the slopes of which increase with increasing molecular length (from Asp to amino-pimelic acid).

The agreement between the adsorption model (Eq. [1]) and the experimental data is evident from the linearity of the plots in Fig. 1. Thus the Langmuir-type isotherm is a reasonable model for the treatment of the present adsorption data. In these plots, points at high concentrations of Asp, Glu, and arginine were not included (entries with asterisks in the tables). These points correspond to apparently anomalous behavior which will be dealt with in the Discussion. Since, according to Eq. [1], the slopes of the lines in Fig. 1 correspond to the reciprocal of the maximum number of adsorption sites, it can be concluded that the adsorption capacity (mole basis) of HA–M increases with decreasing molecular length; the adsorption affinities were calculated from the intercepts of the lines in Fig. 1. A summary of these calculations is shown in Table IX.

According to the figures in column 2 of Table IX, the adsorption affinity of the four acidic amino acids is similar, which suggests that their adsorption involves the same kind of sites on the adsorbent and the same functional group in the adsorbates. The affinities of succinic and adipic acids, however, are substantially higher than those of their amino acid homologs; an explanation for these results is advanced in the Discussion. Whereas the affinities of the acidic amino acids are very similar, the figures in column 3 of Table IX show a marked decrease in the maximum number of adsorption sites with the addition of each methylene group; in fact, addition of a $-CH_2$-group to a molecule reduces the apparent number of sites to approximately $\frac{1}{2}$. Also, it may be seen that the number of sites for adipic acid is similar to the number of sites for its homolog, amino-adipic acid; in the case of succinic acid there is a substantial difference in the number of adsorption sites when compared to the value for aspartic acid. It is important to realize that the magnitude of adsorption depends on both the parameters K and N, as shown in Table IX.

TABLE VI

Adsorption Isotherm of Adipic Acid on HA–M at 37°C

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
2.28	0.435	1.66
4.56	0.904	3.09
6.78	1.02	3.43
6.83	1.41	4.31
13.7	2.20	5.34
17.1	2.82	5.99

Lack of attention to this point may lead to erroneous conclusions. For example, a comparison of columns 2 and 3 in Tables I and II indicates, in a qualitative fashion, that, for similar equilibrium concentrations (column 2), larger amounts of Asp than of Glu were adsorbed onto the HA–M. However, according to the adsorption model, both amino acids have the same affinity; therefore, the apparent higher adsorption for Asp is related to the higher number of adsorption sites for this amino acid, as shown in Table IX. In fact, if the adsorption values for the various adsorbates are normalized with respect to the maximum number of sites and the resulting figures are plotted against the equilibrium concentration, as shown in Fig. 2, it is clear that the individual isotherms for the acidic amino acids coalesce into a single plot; a small deviation is observed to the amino-pimelic acid. This result is in agreement with expectation since in this kind of plot, resolution of the curves should reflect only differences in affinities (parameter K in Table IX). The similarities and differences displayed by the figures in Table IX are further explained by a mechanistic model in the Discussion.

The properties of HA–M as an adsorbent were significantly modified by the pretreatments described under Materials and Methods (pertinent analytical data shown in Tables VII and VIII). It is seen in Table IX that pretreatment of HA–M with a $CaCl_2$ solution resulted in a considerable increase in the maximum number of adsorption sites for Asp. A more striking result was obtained with L-arginine; in this case, no detectable adsorption occurred with the untreated hydroxyapatite, whereas a well-defined adsorption isotherm was obtained with a sample of HA–M pretreated with the phosphate solution at pH 7.8. Although the number of sites was smaller by comparison with the acidic amino acids, the adsorption affinity is very high. It is pertinent to point out that pretreatment of the HA with the phosphate solution at pH 6.8 resulted in adsorption values which were only about one-half of those obtained with the phosphate solution at pH 7.8. However, a complete isotherm for the pH 6.8 pretreatment was not produced.

The adsorption isotherm for Asp on HA–L is shown in Fig. 3. In this case, a smooth plot was obtained, even for concentration regions beyond those used with HA–M. In the latter case, an anomalous behavior was observed (entries with asterisk in Table I) when the equilib-

TABLE VII

Adsorption Isotherm of L-Aspartic Acid on HA–M Pretreated with $CaCl_2$ at 37°C[a]

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
7.51	0.488	5.27
15.0	0.982	10.4
22.5	1.57	13.8
30.1	2.10	18.2
18.8	1.95	18.1
22.6	2.45	20.6
26.3	2.91	23.5
30.1	3.35	26.6
33.8	3.91	28.5
37.6	4.53	29.8
75.1*	5.72	35.9

[a] Entry excluded from linear regression of Eq. [1] indicated by asterisk.

TABLE VIII

Adsorption Isotherm of L-Arginine on HA–M Pretreated with Potassium Phosphate pH = 7.8 at 37°C[a]

Initial concentration (μM/ml/g)	Equil. concentration (μM/ml)	Adsorption (μM/g)
0.543	0.0861	0.274
1.44	0.207	0.438
1.33	0.207	0.411
1.42	0.241	0.455
1.40	0.240	0.447
0.959	0.224	0.422
2.62	0.522	0.522
1.88	0.494	0.526
2.88*	0.709	1.02
3.82*	0.899	1.66
4.72*	1.11	2.16

[a] Entries excluded from linear regression of Eq. [1] indicated by asterisks.

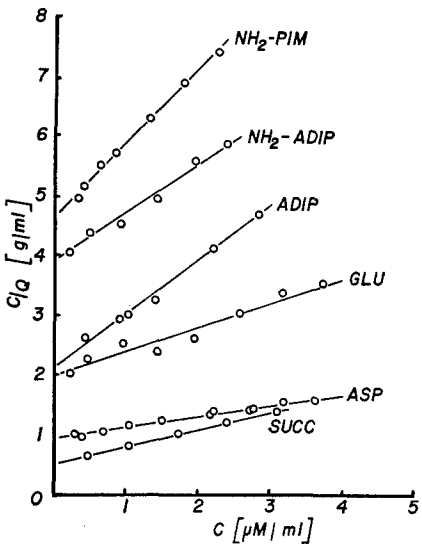

FIG. 1. Adsorption isotherms plotted according to Eq. [1].

rium concentration was higher than about 3.4 μM/ml.

DISCUSSION

The results of this investigation show that a relatively simple adsorption model fits the experimental measurements to a satisfactory degree (Fig. 1). This adsorption model also constitutes a valuable tool for elucidating the mechanisms involved in the adsorption process. Of particular significance is the fact that the affinity parameters for the adsorption of the four acidic amino acids tested are very similar (Table IX), which may be interpreted as indicating that similar functional groups in the amino acids and at the adsorption sites in the apatite are involved in the adsorption phenomenon. It is reasonable to identify the functional groups as the terminal carboxyls in the adsorbates, and the adsorption sites as Ca^{2+} ions on the surfaces of the HA crystals. The maximum number of adsorption sites did decrease, in a systematic fashion, with increasing length (or molecular weight) of the adsorbed molecule; this observation probably reflects chemical and geometrical properties of the adsorption process which will be discussed below.

An agreement, such as that found in the present work, between an adsorption model and the experimental measurements does not imply that the actual molecular adsorption follows the restrictions imposed by the selected model [(a) the adsorption sites are isoenergetic, (b) only a monolayer of adsorbate is formed, and (c) there are no interactions between the adsorbate molecules either in the solution or on the surface of the adsorbent]. Nevertheless, the more the experimental observations can be explained by a given model, the more likely it is that the model describes the actual adsorption process. The model used in the present in-

TABLE IX

Adsorbate		Affinity [ml/μM] (*K* in Eq. [1])	Max. number of sites [μM/g] (*N* in Eq. [1])	Correlation coefficient[a]
L-Aspartic acid		0.203	53.5	0.984
L-Glutamic acid		0.206	24.4	0.962
DL-α-amino adipic acid		0.218	11.9	0.986
DL-α-amino pimelic acid		0.275	7.85	0.998
Succinic acid		0.558	35.3	0.999
Adipic acid		0.425	11.1	0.992
	Pretreated			
L-Aspartic acid	$CaCl_2$	0.164	72.1	0.978
L-Arginine	phosphate buffer pH = 7.8	9.47	0.635	0.998

[a] Correlation coefficient of linear regression of Eq. [1].

vestigation presupposes the presence of discrete adsorption sites on the HA surfaces. We adopt here the view that these sites correspond to Ca^{2+} ions in the [100] planes of the hydroxyapatite crystallites. This assumption is justified by the crystal habit of HA, namely, thin hexagonal needles with their long axis parallel to the *c* crystallographic axis. Thus, most of the exposed surface area corresponds to [100] planes. The Ca^{2+} ions in these planes are then the Ca(I) ions in the structure (14) which are stacked in a columnar fashion with an interatomic distance of 3.441 Å. The distance between the calcium rows corresponds to length of the *a* parameter of the unit cell, i.e., 9.432 Å; thus, the interatomic distances and the distance between the calcium rows define the geometry of the HA for adsorption purposes. Concerning the geometry of the adsorbate molecules, a recent report (15) gives the interatomic distances and bond angles for Asp and Glu residues. From these figures, it is possible to ascertain the maximum molecular lengths and diameters for the two amino acids. It is also possible, on the basis of the same report, to build up the structure for the other members of the amino acid series studied here, maintaining the structural features apparent in the first two members of the series. A model based on the assumption that the long axis of the acid molecules is either parallel or perpendicular to the surface of the HA crystal would not explain satisfactorily the adsorption data. Thus, we

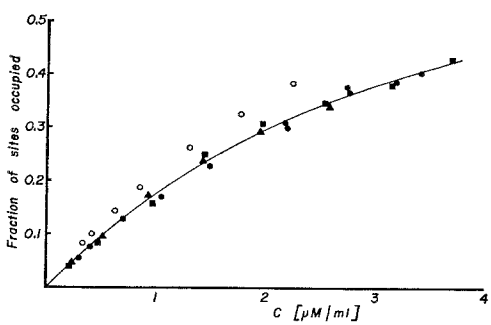

FIG. 2. Fraction of adsorption sites occupied by adsorbate molecules plotted as a function of the concentration at equilibrium. L-Asp, ●; L-Glu, ■; DL-α-amino-adipic, ▲; DL-α-amino-pimelic, ○.

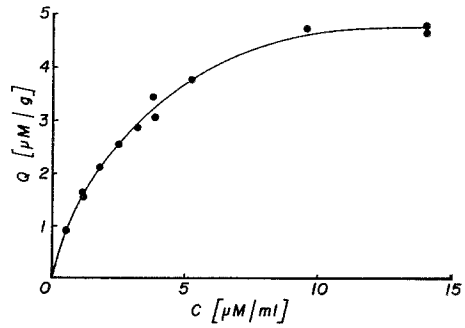

FIG. 3. Adsorption isotherm for L-Asp on HA–L.

have assumed that the amino acid molecules are anchored to the HA by their terminal carboxyl groups with an inclination relative to the crystallite surface. We also consider that the adsorbed molecule has freedom of rotation; thus, the effective volume of adsorption is described by a cone at the vertex of which there is the terminal carboxyl attached to a calcium ion belonging to the HA surface. Then, any calcium ion located within the projection of the cone on the surface could not function as an adsorption site. If we ascribe an angle of 55° for Asp and an angle of 40° for the rest of the molecules, the foregoing geometrical model would explain the variation in the number of sites with increasing molecular length, as shown in Table X. The projection of the cone generated by Asp yields a circle of radius 3.32 Å; thus, according to the assumptions made, one molecule of Asp could be adsorbed on every other Ca ion in all rows. The projection radius for Glu is 5.23 Å; which results in adsorption every fourth calcium in every row; therefore, the maximum number of adsorption sites for Glu should be one-half of that for Asp. Similar considerations show that the available sites for adsorption of amino-adipic acid would be every fourth calcium in every other row, yielding a maximum number of sites which is one-half of the sites for Glu. Also, on geometrical grounds, the maximum number of adsorption sites for pimelic acid should be smaller than that for amino-adipic acid by 25%.

Considering the uncertainties inherent in the assumptions made, the agreement shown in

TABLE X

Maximum Number of Adsorption Sites Obtained from Eq. [1] and from Geometrical Model

Adsorbate	Max. molecular length (Å)	Max. number of adsorption sites	
		Eq. [1]	Geometric model
L-Aspartic	5.78	53.5	(53.5)[a]
L-Glutamic	6.83	24.4	26.6
DL-Amino adipic	8.31	11.9	13.3
DL-α-Amino pimelic	9.36	7.9	10.0
Succinic	5.78	35.3	35.7
Adipic	8.05	11.1	13.3

[a] Taken as the basis for calculation of the rest of the entries in this column.

Table X between the maximum numbers of sites obtained from Eq. [1] and from geometrical restrictions is satisfactory for the acidic amino acids and the two dicarboxylic acids; thus, the idea of geometrical hindrances yields a model that is in agreement with the parameters obtained from Eq. [1]. The fact that the affinities (parameter K) were the same in the homologous series of acidic amino acids indicates that there is no measurable increase in the work of adsorption when additional CH_2 groups are added to the adsorbate molecule. This result is quite different from that obtained, for example, in the adsorption of the series formic to butyric acid on carbon (16); in this case the affinities increase with each CH_2 group added. Thus, the present results are consistent with the assumption that only the terminal carboxyl is involved in the adsorption of the acidic amino acids. Furthermore, it is reasonable to think that the flexibility of the adsorbed molecule should increase with increasing chain length; therefore, for practical purposes, one would expect a smaller inclination for Asp than for the rest of the members in the series. For the case of succinic acid, however, an inclination of 40° was used rather than 55°; a partial justification for the smaller inclination is that the absence of the α-NH_2 group may enhance electrostatic attraction between the "nonadsorbed" carboxyl and the Ca ions in the adsorbing matrix.

In spite of the consistency obtained with the adsorption model of Eq. [1], certain features demand caution in accepting its validity. As mentioned previously (Results) the Asp and Glu isotherms deviate upward from the Langmuir isotherm at adsorbate concentrations higher than $\approx 3.4\ \mu$M/ml. This anomaly is probably not related to multilayer adsorption because (a) the knee region before the rise corresponds, approximately, to a coverage of only one-half of the maximum number of adsorption sites as derived from Eq. [1]; and (b) the BET model (17) (using reduced concentrations) does not fit the experimental data beyond the knee region of the isotherms even if the number of adsorbed layers in the BET equation was limited to 2 or 3. The increased adsorption at high adsorbate concentrations does not appear to arise from interactions of the adsorbate molecules in solution because the phenomenon was not observed in the adsorption of Asp onto HA–L. A plausible explanation for the anomalous behavior is inhomogeneity of the HA–M; if such is the case, then the requirement of isoenergetic adsorption sites, implied in Eq. [1], would be violated. Nevertheless, it appears that the information and interpretation derived from Eq. [1] using experimental points in the low concentration region of the isotherms are valid; if the adsorption of the various compounds is normalized with respect to the maximum number of adsorption sites, N, and the resulting quantities (Q/N) are plotted against equilibrium concentration in solution, all the points describe the same function (with the possible exception of amino-pimelic acid); see Fig. 2. It is clear that, in this case, the function plotted corresponds to the fraction of sites occupied by the adsorbate at a given equilibrium concentration. The plot in Fig. 2 indicates that the adsorption of all the molecules involves the same energy levels and that the adsorption differences between individual members of the series are simply due to differences in availability of adsorption sites, which appear to be dictated by geometrical considerations.

The maximum possible coverages calculated on the basis of the parameters N are low when

compared to the specific surface area of the HA–M as determined by nitrogen adsorption. Thus, for Asp the maximum number of adsorbed molecules per gram would be given by 53.5×10^{-6} moles/gHA $\times$ 6.022×10^{23} molecules/mole = 3.22×10^{19}. Since this amino acid is effectively occupying one *a* surface of the HA unit cell, the effective coverage per molecule is 64.91 $Å^2$, and therefore, the maximum coverage would be 20.8 m^2/g as compared to 59.56 m^2/g obtained by nitrogen adsorption. There are at least two factors which may contribute to this apparent discrepancy. First, the access of the molecules to the adsorbent surface (interstices and defects) is most probably different for nitrogen than for the molecules studied here. Second, although the assumption has been made that the exposed crystallite surfaces correspond essentially to [100] planes, in reality this may not be completely so since the surfaces of crystals usually display irregularities, dislocations, etc. (18), which may contribute to a lower adsorption than anticipated. In addition, even in the [100] planes there are probably vacancies in the calcium sites. In fact, an increased adsorption of Asp was observed when the HA–M was pretreated with a $CaCl_2$ solution; in this case the calculated maximum coverage amounted to 28.6 m^2/g. The result that L-arginine is not adsorbed unless the HA is pretreated with a phosphate-containing solution also suggests the presence of vacancies (or defects) at the crystallite surface. However, nonspecific calcium or phosphate enrichment (unrelated to vacancies) of the surface would also be consistent with the results obtained.

The affinities obtained for succinic and adipic acids (Table IX) are significantly higher than those for the acidic amino acids. In the former cases, there is a complete equivalence of the two carboxyl groups; hence, the probability of forming a stable bond with the surface calcium ions should be higher than in the case of the amino acids. It is pertinent to point out that the lengths of the two dicarboxylic acids do not match distances between two calcium ions on the apatitic surface. In fact, with the exception of Glu, this kind of mismatch is common to all the adsorbates studied. This observation is consistent with the idea advanced here that molecular adsorption involves only one of the carboxyl groups.

The adsorption of acidic amino acids onto HA mediated by the surface calciums is in agreement with ideas previously advanced (1, 7, 8); the present work substantiates this idea, we believe, in a more definitive way. Barnardi *et al.* (7) observed that arginine was adsorbed onto HA to a larger extent in the presence of phosphate buffer at pH 7.8 than at pH 6.8; to some extent, we have verified this finding in that pretreatment of HA–M with phosphate buffer at pH 7.8 produced adsorption values which were about double the values obtained with pretreatment at pH 6.8 Although we did not obtain enough points to generate a complete isotherm for the latter case, it appears that the increased adsorption is related to an increase in the affinity. A possible explanation for this result could be that at a pH of 7.8, a substantial fraction of the total phosphate ($\approx 80\%$) is in the form of HPO_4^{2-}, whereas at pH 6.8 most of the phosphate ($\approx 70\%$) is in the form of $H_2PO_4^-$; thus, pretreatment of the HA with the buffer at 7.8 could result in enrichment of the HA surface with phosphate ions of a higher valence, thus creating sites with a higher affinity for the positive groups in the adsorbate. It is pertinent to point out that Tiselius *et al.* (1) reported an increase in adsorption affinity of proteins onto HA when the adsorbent was treated with 0.05 *M* $CaCl_2$ solution. This report parallels our finding with acidic amino acids (Asp) and suggests that the adsorption mechanism for proteins may also involve carboxyl or other negatively charged groups in the side chains. The stronger adsorption of proteins as compared to dicarboxylic amino acids (1) might be explained by multiple bonding of caboxyls in the side chains of the proteins; this observation does not invalidate the concept of localized adsorption sites on the HA surface.

The adsorption characteristics of Asp, as compared to the other acidic amino acids tested, would explain the behavior of Asp in thin-layer

adsorption chromatography with HA. Hofmann (19) reported that Asp had the lowest R_F value in a series of 20 amino acids chromatographed. In a chromatographic process, not only is the affinity important, but also the effective number of adsorption sites for the adsorbate. This number of sites is the highest for Asp within the series tested in the present work (Table IX). Thus, qualitatively, there is agreement between the present results and those obtained in related investigations.

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